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LAB 06	
Aim:	Apply the concept of Thread in java.
Theory:	Threads are lightweight subprocesses, representing the smallest unit of execution with separate paths. The main advantage of multiple threads is efficiency (allowing multiple things at the same time). For example, in MS Word, one thread automatically formats the document while another thread is taking user input. Additionally, multithreading ensures quick response, as other threads can continue execution even if one gets stuck, keeping the application responsive. ❖ **Concept of Multitasking** To help users Operating System accommodates users the privilege of multitasking, where users can perform multiple actions simultaneously on the machine. This Multitasking can be enabled in two ways: 1. **Process-Based Multitasking** 1. **Thread-Based Multitasking** **1.Process-Based Multitasking (Multiprocessing)**: In this type of Multitasking, processes are heavyweight and each process was allocated by a separate memory area. And as the process is heavyweight the cost of communication between processes is high and it takes a long time for switching between processes as it involves actions such as loading, saving in registers, updating maps, lists, etc. **2. Thread-Based Multitasking:** As we discussed above Threads are provided with lightweight nature and share the same address space, and the cost of communication between threads is also low. Now, we can understand why threads are being used as they had the advantage of being lightweight and can provide communication between multiple threads at a Low Cost contributing to effective multi-tasking within a shared memory environment. ❖ **Life Cycle of Thread** There are different states Thread transfers into during its lifetime, let us know about those states in the following lines: in its lifetime, a thread undergoes the following states, namely: 1. New State 1. Active State 1. Waiting/Blocked State 1. Timed Waiting State 1. Terminated State

Program:	// sheikh wasimuddin 23071044 cseiot-73
	class MyThread extends Thread {
	public void run() {
	for (int i = 1; i <= 5; i++) {
	System.out.println("Thread running: " + i);
	try {

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```
                Thread.sleep(500); // Pause for 0.5 seconds
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }

    public static void main(String[] args) {
        MyThread t1 = new MyThread();
        t1.start(); // Start the thread
        System.out.println("Main thread finished");
    }
}
```

Output:

thread.java 2

thread.java > Language Support for Java(TM) by Red Hat > MyThread

```
// sheikh wasimuddin 23071044 cseiot-73
class MyThread extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println("Thread running: " + i);
            try {
                Thread.sleep(millis:500); // Pause for 0.5 seconds
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }

    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        MyThread t1 = new MyThread();
        t1.start(); // Start the thread
        System.out.println(x:"Main thread finished");
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\student\Desktop\Iot-73>  & 'C:\Program Files\Java\jdk-22\bin\java.exe
edhat.java\jdt_ws\Iot-73_908e7a1\bin' 'MyThread'
Main thread finished
Thread running: 1
Thread running: 2
Thread running: 3
Thread running: 4
Thread running: 5
PS C:\Users\student\Desktop\Iot-73>
```

Conclusion

Implementing the concept of Thread in java executed successfully.

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